Mateti Karthik

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EDUCATION

Sai University **Aug 2024 – Aug 2028**
B.Tech – Computing & Data Science, CGPA: 8.89/10 *Chennai, Tamil Nadu*

SR Junior College for Boys **Jun 2022 – Apr 2024**
Class XII, Telangana State Board of Intermediate Education, 96.9% *Warangal, Telangana*

EXPERIENCE & PROJECTS

Indian Institute of Information Technology Pune **Jun 2026 – Aug 2026**

Summer Research Intern | Under Dr. Priyank Jain (certificate of completion) **Source Code**

- Conducted research on Federated Learning for Privacy-Preserving Machine Learning, focusing on protecting sensitive data while maintaining model utility.
- Worked on Privacy-Preserving Data Publishing using XGBoost, k-Anonymity, l-diversity, and Differential Privacy.
- Investigated privacy-preserving techniques for structured datasets and analyzed their impact on data utility and machine-learning performance.
- Studied data anonymization and privacy mechanisms for enabling secure use of sensitive data in ML workflows.
- Reviewed research literature on privacy-preserving machine learning and analyzed approaches for real-world applications.

Ideas TIH, Indian Statistical Institute (ISI), Kolkata **May 2026 – Jul 2026**

Summer Intern | Under Dr. Suvra Jyoti Choudhury (certificate of completion) **Source Code**

- Developed a Multi-Agent Retrieval-Augmented Generation (RAG) framework for intelligent resume screening and candidate evaluation.
- Designed specialized LLM agents for resume understanding, skill matching, candidate evaluation, and workflow orchestration using prompt engineering.
- Built an automated candidate-evaluation pipeline integrating retrieval-augmented generation with multi-agent LLM workflows.
- Evaluated the framework on the CareerCorpus dataset and benchmarked its performance against a single-LLM baseline using multiple evaluation metrics.
- Completed training in Statistics, Data Visualization, GitHub, Cloud Computing, Streamlit, Machine Learning, LLMs, Deep Learning, Computer Vision, Agentic AI, Reinforcement Learning, and Data Science systems.
- Completed 7 weeks of individual project work involving data collection, analysis, coding, project discussions, and technical report writing.
- Achieved 29/30 (Outstanding) in the internship project assessment.

Infosys Springboard **Aug 2025 - Oct 2025**

AI/ML Intern (certificate of completion) **Source Code**

- Developed a Live Meeting Summarizer using Speech-to-Text, speaker diarization, Natural Language Processing, and Large Language Models.
- Integrated Whisper/Vosk-based transcription, speaker identification, and automated meeting summarization workflows.
- Built a Streamlit application for real-time transcription, speaker-aware processing, summary generation, and meeting management.
- Integrated Hugging Face models and API-based inference to generate structured meeting summaries.
- Implemented summary export and email delivery functionality for convenient sharing of meeting insights.

TECHNICAL SKILLS

Programming Languages: C, C++, Python, JavaScript

AI & Machine Learning: Machine Learning, Deep Learning, NLP, Computer Vision, Federated Learning, Privacy-Preserving ML, Data Analysis, Explainable AI (XAI)

Generative AI & Agentic AI: LLMs, RAG, Multi-Agent Systems, Agentic AI, Prompt Engineering, AI Agents, Semantic Search, Document Understanding, Research Automation

Privacy & ML Security: XGBoost, k-Anonymity, l-Diversity, Differential Privacy, Data Anonymization

AI Tools & Frameworks: Streamlit, Hugging Face, Scikit-learn, Pandas, NumPy, Matplotlib, Whisper, Vosk, PyAnnote\

Web & Databases: HTML, CSS, React.js, Node.js, Express.js, MySQL, MongoDB

Developer Tools: Git, GitHub, API Integration

ACHIEVEMENTS

- Semi-Finalist, Convolve 4.0 – Generative AI Track: Selected among 8,500+ participants across 40+ institutions in a national-level AI/ML hackathon powered by IDFC FIRST Bank.
- Top 20 Finalist, SPARQ 2026 (Senior Category): Ranked among the Top 20 participants in Statistical and Probabilistic Analysis of Real-Life Quantitative Problems and qualified for the Industry Interaction Camp.
- Final Selection Round, BodhiTree Summer Internship Program – IIT Bombay.